

ILAM, Iran, Islamic Republic of

WMO: **407800**

Lat: **33.5884N** Lon: **46.3972E** Elev: **4386** StdP: **12.51** Time Zone: **3.50 (IRN)** Period: **02-19** WBAN: **99999**

Annual Heating, Humidification, and Ventilation Design Conditions

	Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF	
				99.6%			99%			0.4%		1%					
		99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD		
	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(1)	1	24.3	27.0	5.3	8.5	44.0	10.0	10.8	44.7	20.2	45.5	17.6	46.5	3.2	100	0.467	(1)

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

	Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB		
			0.4%		1%		2%		0.4%		1%		2%				
			DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD	
	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(2)	7	26.3	102.2	62.5	100.3	62.2	98.5	61.6	64.7	98.1	63.5	96.6	62.5	95.0	9.3	270	(2)

	Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB	
	0.4%			1%			2%			0.4%		1%		2%			
	DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB		
	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(3)	53.9	72.9	62.4	52.1	68.1	62.3	50.3	63.6	63.0	32.1	98.4	31.2	97.0	30.2	95.0	75.7	(3)

Extreme Annual Design Conditions

	Extreme Annual WS				Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
					Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
	1%	2.5%	5%		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
	(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(4)	18.6	16.2	14.0	DB	18.0	104.9	4.7	2.2	14.6	106.5	11.8	107.9	9.2	109.1	5.8	110.7	(4)
(5)				WB	15.7	67.2	4.7	2.4	12.3	69.0	9.6	70.4	6.9	71.8	3.5	73.5	(5)

Monthly Climatic Design Conditions

				Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
				(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6)	Temperatures, Degree-Days and Degree-Hours		DBAvg	62.7	40.4	43.3	50.6	58.5	69.0	79.4	84.9	84.5	76.8	66.4	51.9	44.8	(6)
(7)			DBStd	16.66	5.81	6.50	6.85	5.91	5.64	4.24	3.74	3.36	4.91	5.84	6.08	5.52	(7)
(8)			HDD50	805	299	200	72	5	0	0	0	0	0	0	51	178	(8)
(9)			HDD65	3131	763	608	449	209	26	0	0	0	0	56	395	625	(9)
(10)			CDD50	5424	1	12	92	260	590	881	1083	1070	803	508	106	18	(10)
(11)			CDD65	2274	0	0	3	14	151	431	618	605	353	99	0	0	(11)
(12)			CDH74	31605	0	0	29	178	1883	6048	9182	8736	4476	1067	5	1	(12)
(13)			CDH80	18013	0	0	1	35	655	3445	5762	5472	2367	276	0	0	(13)
(14)	Wind		WSAvg	6.0	5.3	5.6	6.2	6.1	6.1	7.4	7.1	6.9	6.2	5.6	4.8	5.0	(14)
(15)	Precipitation		PrecAvg	19.3	4.1	2.8	3.0	2.2	0.5	0.0	0.0	0.0	0.0	1.0	2.7	2.8	(15)
(16)			PrecMax	31.9	7.1	7.6	7.6	5.0	2.5	0.1	0.3	0.7	0.7	7.5	13.9	6.6	(16)
(17)			PrecMin	10.8	0.7	0.5	0.0	0.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	(17)
(18)			PrecStd	5.0	1.7	1.7	2.2	1.5	0.7	0.0	0.0	0.1	0.1	1.5	2.8	1.8	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB		60.6	65.6	77.4	84.6	92.8	100.8	104.4	103.9	98.9	89.2	73.7	68.3	(19)
(20)			MCWB		43.3	47.9	51.7	56.4	59.5	62.3	63.4	63.2	62.5	58.3	50.8	47.0	(20)
(21)		2%	DB		57.1	61.2	71.9	78.8	89.6	98.5	102.5	102.1	96.7	85.9	70.9	62.8	(21)
(22)			MCWB		42.3	44.8	49.6	55.1	58.1	61.2	62.8	62.2	60.9	56.3	49.9	45.2	(22)
(23)		5%	DB		53.8	58.7	67.7	74.9	86.3	96.5	100.6	100.3	93.7	82.8	67.7	58.8	(23)
(24)			MCWB		41.6	43.9	48.2	53.6	57.0	60.4	62.4	62.0	59.5	55.0	49.2	44.2	(24)
(25)		10%	DB		51.4	55.2	64.2	71.6	83.8	94.0	98.6	98.4	91.5	80.5	64.1	55.1	(25)
(26)			MCWB		41.0	43.2	47.0	51.8	56.6	59.5	61.8	61.6	58.4	54.2	48.8	43.2	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB		48.7	51.2	53.7	58.8	61.8	64.1	66.5	66.1	65.0	60.3	56.4	52.1	(27)
(28)			MCDB		52.7	58.6	66.7	74.8	83.3	97.3	101.0	99.7	93.9	80.4	63.8	56.8	(28)
(29)		2%	WB		46.0	48.6	51.9	57.0	60.2	62.7	64.7	64.6	62.1	58.3	54.2	48.7	(29)
(30)			MCDB		50.5	55.6	65.4	72.6	82.7	95.2	98.9	98.1	92.4	77.0	61.1	56.1	(30)
(31)		5%	WB		44.0	46.0	50.5	55.4	59.0	61.6	63.6	63.5	60.6	57.0	52.7	46.7	(31)
(32)			MCDB		49.9	55.0	63.5	69.2	81.1	93.7	97.2	96.7	91.1	75.2	60.4	54.7	(32)
(33)		10%	WB		42.2	44.4	48.9	54.0	57.8	60.2	62.4	62.4	59.2	56.0	50.9	45.3	(33)
(34)			MCDB		49.2	53.1	60.9	67.1	79.1	91.9	95.8	95.2	89.0	74.4	59.8	52.6	(34)
(35)	Mean Daily Temperature Range		MDBR		18.0	18.5	20.4	21.4	24.0	27.0	26.3	26.8	26.9	23.2	20.1	19.1	(35)
(36)		5% DB	MCDBR		21.7	22.9	24.5	25.4	27.4	29.4	27.5	27.8	27.6	25.9	24.9	24.6	(36)
(37)			MCWBR		12.4	12.1	11.6	11.3	11.1	12.0	11.0	10.9	11.9	11.6	12.4	13.4	(37)
(38)		5% WB	MCDBR		16.7	18.9	19.4	21.1	24.0	27.9	26.6	26.8	26.4	21.1	16.5	17.9	(38)
(39)			MCWBR		11.2	11.6	10.6	10.7	10.2	12.0	11.4	11.5	12.1	10.3	9.5	11.4	(39)
(40)	Clear-Sky Solar Irradiance	taub			0.334	0.373	0.427	0.511	0.501	0.440	0.488	0.445	0.392	0.426	0.363	0.339	(40)
(41)		taud			2.238	2.114	1.955	1.786	1.827	1.969	1.849	1.961	2.094	1.997	2.171	2.233	(41)
(42)		Ebn at Noon			274	273	267	250	254	269	255	265	274	251	259	264	(42)
(43)		Edh at Noon			35	44	56	69	67	58	65	57	48	49	37	33	(43)
(44)	All-Sky Solar Radiation	RadAvg			910	1197	1629	1889	2245	2644	2504	2336	2018	1454	1064	861	(44)
(45)		RadStd			58	89	112	149	127	76	73	70	71	113	109	72	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (3 neighbors)	N/A	N/A	-6.26	+1.01	-3.02	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page